Mohamed Moheyeldin

Forward Deployed Engineer

Reston, VA 20191 | mohamedmoheyeldin.com | mohamedmoheyeldin.jobs@gmail.com | linkedin.com/in/moheyeldin | github.com/mohamedmoheyeldin

Professional profile

Forward Deployed Engineer on a VA healthcare claims platform, with 11 years in software development and quality engineering across federal healthcare, banking, and e-commerce. Integrate health-provider claim-status APIs, write React features, and built Data Generator & File Processing, which replaced hours of daily test-data prep with a 5-minute run. Work directly with stakeholders from first requirements conversations through demos and adoption.

Bring a customer-facing engineering approach to any industry, with the deepest experience in federal healthcare, where production data is off-limits and environments are restricted. Hands-on with React, TypeScript, JavaScript, Java, SQL, REST APIs, Tableau, and CI workflows, and with generative AI: Claude API, OpenAI and Azure OpenAI, LangChain, LlamaIndex, embeddings, vector databases, prompt engineering, agentic workflows with MCP, and RAG evaluation for answer accuracy and hallucinations. Built Claims Document Assistant, a Claude-based claims PDF assistant with structured outputs, rule-based checks, human review, and an evaluation harness scored against labelled ground truth.

Professional experience

Booz Allen Hamilton — Washington, DC

Forward Deployed Engineer (FDE) | Dec 2020 – Present

Customer: U.S. Department of Veterans Affairs (VA)

Work directly with VA stakeholders on a healthcare claims platform with two frontends sharing a database, supporting claim submission, provider updates, claim progress and status tracking, and payment visibility.

- Work with health providers to integrate APIs with the VA system so live claim statuses display for VA staff, providers, and veterans based on their role; the VA side previously had no continuously updated claim status.
- Responded to stakeholder questions about stalled claims and process gaps by creating a claims framework that notifies the right party when a claim needs extra work, keeping claims moving and giving better tracking of every claim.
- Built and continue to enhance Data Generator & File Processing because production data can't be used in VA lower environments. The team used to spend hours each day preparing data and keeping a database in every environment; now anyone can generate a full development database or up to 10,000 custom records in under 5 minutes, and the whole team uses it daily.
- Run the tool like a product: versioned Windows releases built through GitHub Actions, in-app release notes, a user guide, a stakeholder overview, and release updates for the team; added Excel file processing with duplicate detection as a second workflow when team needs grew.
- Enabled the team to use Tableau for requests and system information so any user can build their own dashboard; built custom dashboards for VA and internal users and, through repeated VA stakeholder discussions, expanded Tableau into the main way to monitor server health and system status.
- Contribute React, TypeScript, and JavaScript features that turn stakeholder feedback into improvements to the claims application.
- Work directly with VA stakeholders to map healthcare claims workflows, clarify technical and operational constraints, and define requirements and acceptance criteria for claim submission, provider updates, status tracking, and payment visibility.
- Lead the stakeholder demonstrations at the end of each release cycle and program increment (since 2022) and all discussions with VA stakeholders; gather feedback on daily workflows and refine implementations around user needs.
- Join working sessions with health-provider claims processors and, at times, VA end users on issues reported in production, to see the full picture and keep future work pointed in the right direction.
- Investigate defects and data discrepancies across frontend, backend, and MySQL data; coordinate with developers and stakeholders to resolve delivery dependencies, and document progress, blockers, and operational impact in Jira.
- Evaluated alternatives to legacy automation and helped select Playwright, weighing maintainability, reuse, authentication, and restricted environment access; developed a reusable smoke and regression framework that helps QA and developers reduce production bugs and deliver stakeholder requests faster.
- Apply prompt engineering with GitHub Copilot in VS Code for development, debugging, refactoring, and test creation; use Gemini, GPT models, and Claude to compare implementation approaches, and review and test every generated change before it ships.

Professional experience continued

Chick-fil-A Corporate — Atlanta, GA

Software Engineer in Test | Nov 2018 – Dec 2020

Built reusable test tooling and used it to investigate REST APIs, authentication, and AWS test environments, giving engineers the evidence to track down failures.

- Built reusable JavaScript browser automation around application workflows, translating expected behavior into repeatable checks that supported implementation and release decisions.
- Investigated application-to-service integration through browser automation and Postman REST API checks, validating schemas, service contracts, and data-dependent behavior to surface discrepancies.
- Validated OIDC, Okta MFA, and SSO workflows, troubleshooting authentication behavior across application scenarios and capturing evidence for failure investigation.
- Connected GitHub Actions and AWS test environments into repeatable delivery workflows, making execution results and diagnostic artifacts available to engineering teams.
- Used AWS EC2, S3, and Aurora resources to support test environments; combined network stubbing, headless testing, and parallel execution to isolate dependencies and reproduce scenarios.

Ally Bank — Sandy, UT

Software QA Engineer | Sep 2015 – Nov 2018

Worked with developers, analysts, and product owners to turn banking requirements into traceable tests and release checks.

- Translated business and functional requirements into test plans, acceptance criteria, and traceable scenarios with developers, analysts, and product owners.
- Performed exploratory, functional, and regression testing of banking web applications, investigating behavior across customer workflows and platforms.
- Created reproducible Jira defect reports with environment details and business impact; collaborated with developers to investigate causes and verify fixes.
- Used SQL data checks alongside application validation to investigate discrepancies and verify expected results.
- Supported smoke, regression, and sanity checks across development, QA, UAT, and staging environments, including deployment checks and hotfix verification.

Selected projects

Claims Document Assistant: Claude API assistant: prompt-engineered extraction of 17 claim-PDF fields, rule checks, human review, and an eval harness. github.com/mohamedmoheyeldin/claims-document-assistant

Portfolio & Career Content System: Built a React/Untitled UI portfolio from validated JSON, with shared PDF/Word generation and static deployment packaging for Cloudflare. mohamedmoheyeldin.com

Technical skills

Application development: React, TypeScript, JavaScript, Java, Reusable components

Data and integration: SQL, MySQL, JSON, REST APIs, Postman, Tableau, SQL Server, Oracle, MongoDB

Delivery and quality: Playwright, GitHub Actions, Git, Jira, Jenkins, AWS EC2/S3/Aurora test environments

Generative AI & agents: Claude API, OpenAI API, Azure OpenAI, LangChain, LlamaIndex, RAG, Embeddings, Vector databases, Prompt engineering, RAG evaluation (accuracy, hallucinations), Agentic AI, MCP, Claude Code, GitHub Copilot, VS Code, GPT models, Claude, Gemini

Education

Bachelor's degree in Computer Science | American College of Commerce and Technology | May 2014

Professional development

Claude Academy | Issued 2026: Introduction to Claude Cowork; Introduction to Claude Tag; AI Capabilities and Limitations; Introduction to Model Context Protocol; Claude Code in Action; Building with the Claude API

OpenAI Academy | Issued 2026: AI Foundations; Agents and Workflows; Applied AI Foundations; AI Leadership; Apply AI at Work Pathway Completion

Palantir | Issued 2026: Speedrun: Your First End-to-End Workflow; Foundry & AIP Builder Foundations Quiz

Scaled Agile | Issued 2021: Implementing SAFe Course (5.1)